# SIMATS ENGINEERING

## ASSESSMENT TOOL 1

**COURSE CODE:** CSA5116
**COURSE NAME:** Cryptography and Network Security

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**COURSE OUTCOME COVERED**

**CO2:** Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Assessment Tool Weightage: **40%**

**QUESTIONS:** **Total Marks: 50**

**Annexure A**

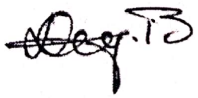
## Analytical Problem Solving

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***Code of Conduct:***

*I Deepak Kumar B/ 192424227_certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***



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5. ECC vs
Given:

## Annexure A Analytical Problem Solving

**1. RSA Encryption Analysis**
Given p=11, q=13

a. Compute n $\phi(n)$\
b. Choose e=7 and find the private key d
c. Encrypt the message M=9
d. Decrypt the ciphertext and verify the result
e. Analyze how changing e affects security

**2. Block Cipher Mode Evaluation**
Given plaintext blocks:
P=[1010,1010,1111,1010], Key = K=1100

a. Encrypt using ECB and CBC modes (Assume IV = 0000)
b. Compare the ciphertext outputs
c. Analyze which mode hides patterns better and explain why

**3. DES vs 3-DES vs Blowfish Performance Study**
Given:

a. DES key size = 56 bits, time = 2 ms/block
b. 3-DES key size = 168 bits, time = 6 ms/block
c. Blowfish key size = 128 bits, time = 1 ms/block

For 1000 blocks of data:

i.Calculate total encryption time for each algorithm ii.Analyze trade-offs between security and performance
iii.Recommend the best algorithm for secure real-time communication

**4. Diffie-Hellman Key Exchange Analysis**
Given:

a. p=23, g=5
b. User A private key = 6
c. User B private key = 15
d. Compute public keys for A and B
e. Compute the shared secret key
f. Demonstrate how a Man-in-the-Middle attacker could alter the exchange
g. Suggest a secure method to prevent the attack

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**5. ECC vs RSA Comparison**

*Given:*

a. RSA key size = 1024 bits
b. ECC key size = 160 bits (equivalent security)
c. Device processing capacity = limited (IoT device)
d. Analyze computational cost for both systems
e. Estimate which consumes less power and memory
f. Justify which cryptosystem is better for the given scenario with reasoning

**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Problem Understanding | 20% | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection | 20% | Selects most appropriate method/formula with strong justification | Selects correct method with minor errors | Inappropriate or partially correct method | Unable to select method |
| Solution Process | 25% | Logical, systematic steps with correct approach throughout | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors | No clear process |
| Accuracy of Calculation | 20% | All calculations accurate; correct final answer | Minor calculation errors | Frequent errors; incorrect final answer | No valid calculations |
| Interpretation of Results | 15% | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation | No interpretation |

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REG NO: 19242227

COURSE CODE: CSA 5116 - Cryptography and Network Security.

2. Block Cipher Mode Evaluation:

Given:

* Plain text blocks

$$P = [1010, 1010, 1111, 1010]$$

* $K = 1100$
* Initialization Vector: $IV = 0000$

a) Encryption using ECB and CBC modes

1. ECB (Electronic Codebook)

Formula:

$$C_i = E_K(P_i) = P_i \oplus K$$

| Block | Plaintext | key | Cipher text |
|---|---|---|---|
| 1 | 1010 | 1100 | 0110 |
| 2 | 1010 | 1100 | 0110 |
| 3 | 1111 | 1100 | 0011 |
| 4 | 1010 | 1100 | 0110 |

ECB cipher text: [0110, 0110, 0011, 0110]

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2. CBC (Cipher Block Chaining)

Formula:

$$C_i = E_k (P_i \oplus C_{i-1})$$

$$C_0 = IV = 0000$$

| Block | calculation | Ciphertext |
|---|---|---|
| 1 | $1010 \oplus 0000 \oplus 1100$ | 0110 |
| 2 | $1010 \oplus 0110 \oplus 1100$ | 0000 |
| 3 | $1111 \oplus 0000 \oplus 1100$ | 0011 |
| 4 | $1010 \oplus 0011 \oplus 1100$ | 0101 |

CBC ciphertext [0110, 0000, 0011, 0101]

b) Compare ciphertext outputs

ECB mode

0110, 0110, 0011, 0110

Repeated plaintext produces repeated ciphertext

Each block is encrypted independently

CBC Mode

0110, 0000, 0011, 0101

Repeated plaintext can produce different ciphertext.

Each block depends on the previous ciphertext block.

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c.) Which mode hides patterns better?

CBC hides patterns better than ECB because each plaintext is XORed with the previous ciphertext block before encryption. Therefore identical plaintext blocks can produce different ciphertext blocks, making patterns harder to identify.

ECB reveals repeated patterns, whereas CBC provides better pattern concealment when used correctly with a suitable I.V.

3. DES Vs 3-DES vs Blowfish performance study.

| Algorithm | Key size | Encryption time per block |
|---|---|---|
| DES | 56 bits | 2ms |
| 3-DES | 168 bits | 6ms |
| Blowfish | 128 bits | 1ms |

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i) Calculate total encryption time for 1000 blocks

Formula

Total time = Time per block * 1000

1. DES

= 2 * 1000 = 2000 ms = 2 seconds

2. 3-DES

= 6 * 1000 = 6000 ms = 6 Seconds

3. Blowfish

= 1 * 1000 = 1000 ms = 1 Second

ii) Analyze trade-offs between security and performance.

* DES : Fast enough for basic encryption but has a small 56-bit key, making it vulnerable to brute-force attacks.

* 3-DES : Provides stronger protection than DES against key-search attacks, but is slower.

* Blowfish : Offers good performance with a variable key length and its faster than the other two in the given data.

blocks

(ii) Recommend the best algorithm for secure real-time communication.

*Recommendation: Blowfish among the three, based on the given timing data.

* Blowfish takes only 1 second to encrypt 1000 blocks, making it the fastest option in this comparison. However, its 64-bit block size is a limitation for modern communication systems.

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